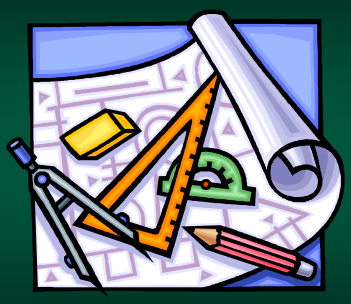
Analysis of Algorithms

Part 1

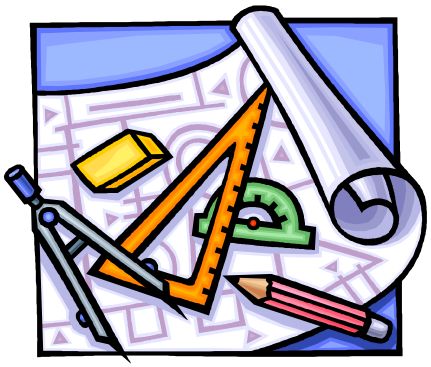


Analysis of Algorithms

Looking at how well it works

What is an Algorithm?

- An *algorithm* is a sequence of unambiguous instructions that solves a problem
- Can be represented various forms – i.e. languages
- Each unique set of data fed into an algorithm specifies an *instance* of that algorithm



Analysis of Algorithms

- Algorithms must to analyzed to determine whether it should be used
- This field is called *algorithmics*
- How it is analyzed:
  - correctness
  - unambiguity
  - effectiveness
  - finiteness/termination - does it in a *finite* amount of time

Correctness

- Correctness means the algorithm obtains the required output with *valid* input
- In other words, does it do what it is supposed to do
- *Proof of Correctness* can be easy for some algorithms – and quite difficult for others
- *Proof of Incorrectness* is quiet easy – find one instance where it fails on valid input

Effectiveness

- How good is the algorithm?
- Two major areas of interest:
  - *time efficiency* defines how long the algorithm will take to complete
  - *space efficiency* defines how much memory and resources will be needed
- … and how these react as the data set grows

Effectiveness

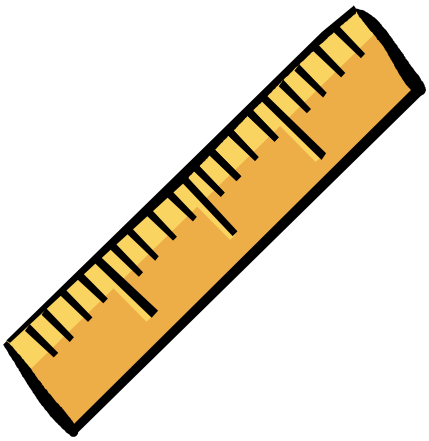
- Knowing this, we can determine if there is a better algorithm
- Does there exist a better algorithm?
  - better time complexity
  - better space efficiency
- *Efficiency is a **HUGE** part of creating professional programs*

Determining Effectiveness

Is the algorithm good?

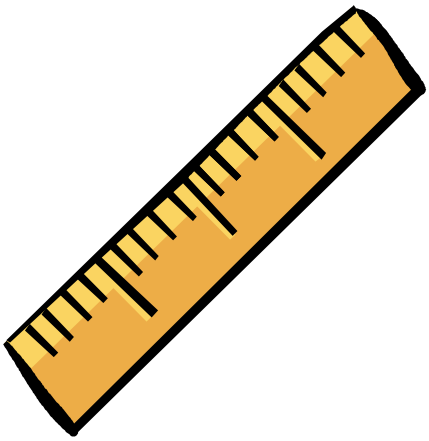
Determining Effectiveness

- Determining if an algorithm is efficient – and will work best to solve a problem – *is vital*
- Some algorithms may work incredibly well
- … and sometimes fail horribly



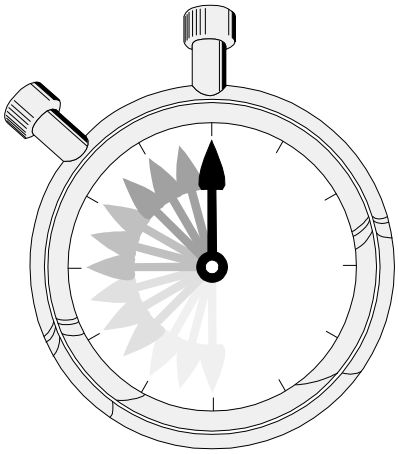
Determining Effectiveness

- Moreover, some algorithms are sensitive to the type of data
- And two algorithms – which solve the same problem – may act *differently* given a set of values



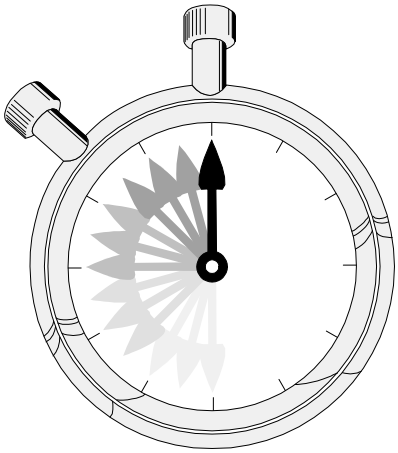
Time Complexity

- Given that computer programs are designed to be fast (and thus efficient), estimating how long an algorithm will take is useful
- *What is the task, repeated by the algorithm, has the most impact on the time?*



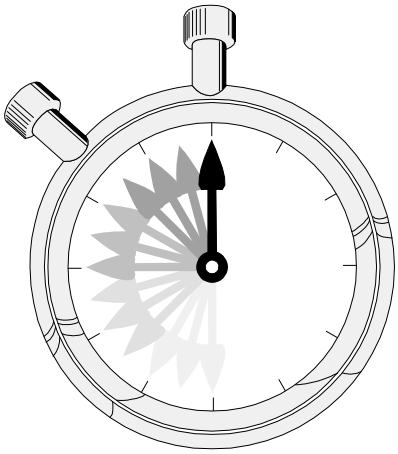
Time Complexity

- The *basic operation* contributes the ***most*** towards the running time of the algorithm
- It is the task that is repeated (often in a loop) by the algorithm



Time Complexity

- We could time the basic operation and then estimate how many times its executed
- This will give a rough idea of the total runtime



Theoretical Analysis of Time Efficiency

Total execution time

Number of times the basic operation executes

$$T(n) \approx c_{op} C(n)$$

number of items in the data set

Cost: execution time of a basic operation

Empirical analysis of time efficiency

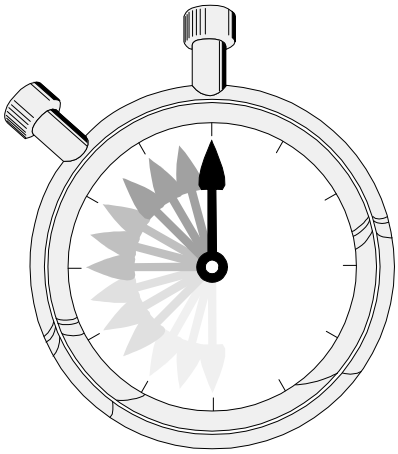
- *Empirical Analysis* can be performed by *observation*
- Select a specific (typical) sample of inputs
- Then physical unit of time and / or count actual number of basic operation's executions
- Analyze the data to <u>estimate</u>: $T(n)$, C_{op}, and $C(n)$

Time Complexity Cases

- Worst case: $C_{worst}(n)$
  - maximum executions over a set of size n
  - can be linear, quadratic, or even exponential!
  - the worst case can be <u>exceedingly</u> rare
- Best case: $C_{best}(n)$
  - minimum executions over a set of size n
  - best case can also be exceedingly rare

Time Complexity Cases

- Average case: $C_{avg}(n)$
  - execute with typical data…
  - …in other words, the type of data the algorithm will it normally encounter
  - this is **NOT** the average of worst and best case



Order of Growth

Uh, "O"

Order of Growth

- For some algorithms, efficiency depends on the *form* of input
- Sometimes, the order of data, or the type of data can *drastically* increase cost

Order of Growth

- In the previous equation, notice that C(n) represents the total number of times the basic operation is executed
- But, how does C react to n?

Order of Growth

- We can analyze n as it approaches ∞
- Examples:
  - how will it run on a computer that is twice as fast?
  - how long does it take with twice the input?

Order of Growth

- One property of functions that we are interested in its rate of growth
- *Rate of growth* doesn't simply mean the "slope" of the line associated with a function
- Instead, it is more like the *curvature* of the line

Several Growth Functions

- There are several functions
- In increasing order of growth, they are:
  - Constant ≈ 1
  - Logarithmic ≈ log n
  - Linear ≈ n
  - Log Linear ≈ n log n
  - Quadratic ≈ n^2
  - Exponential ≈ 2^n

Growth Rates Compared

n =	1	2	4	8	16
1	1	1	1	1	1
log n	0	1	2	3	4
n	1	2	4	8	16
n log n	0	2	8	24	64
n^2	1	4	16	64	256
n^3	1	8	64	512	4096
2^n	2	4	16	256	65536

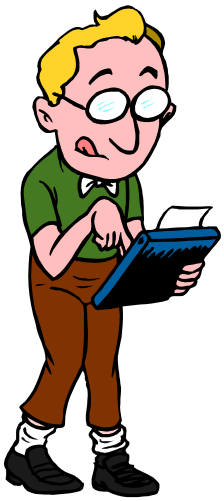
Classifications

- Using the known growth rates…
  - algorithms are classified using three notations
  - these allows you to see, quickly, the advantages/disadvantages of an algorithm
- Major notations:
  - Big-O
  - Big-Theta
  - Big-Omega

Order of Growth

Notation	Name	Meaning
O (*n*)	Big-O	class of functions f(n) that grow no faster than n
Θ (*n*)	Big-Theta	class of functions f(n) that grow at same rate as n
Ω (*n*)	Big-Omega	class of functions f(n) that grow at least as fast as n

Big-O



- So, Big-O notation gives an *upper bound* on growth of an algorithm
- We will use Big-O almost exclusively rather than the other two

Big-O

- The following means that the growth rate of f(n) is no more than the growth rate of n
- This is one of the classifications mentioned earlier

$$f(n) \text{ is } O(n)$$

Why it is O-some!

- These classes make it is easy to…
  - compare algorithms for efficiency
  - making decisions on which algorithm to use
  - determining the scalability of an algorithm
- So, if two algorithms are the same class…
  - they have the same rate of growth
  - both are equally valid solutions

O(1)

- Represents a constant algorithm
- It does not increase / decrease depending on the size of n
- Examples
  - appending to a linked list *(with an end pointer)*
  - array element access
  - practically all simple statements

O(log n)

- Represents logarithmic growth
- These increase with n, but the rate of growth diminishes
- For example: for base 2 logs, the growth only increases by one each time n doubles

O(log n) Examples

- Searching for an item on a sorted array – (e.g. a binary search)
- Traversing a sorted tree

O(n)

- Represents an algorithm that grows linearly with n
- Very common in programming – for iteration
- Examples:
  - finding an item in a linked list
  - merging two sorted arrays

O(n log n)

- Represents an algorithm that has "log linear" growth
- These algorithms grow based on both n and n's log value

O(n log n) Examples

- Quick Sort
- Heap Sort
- Merge Sort
- Fourier transformation

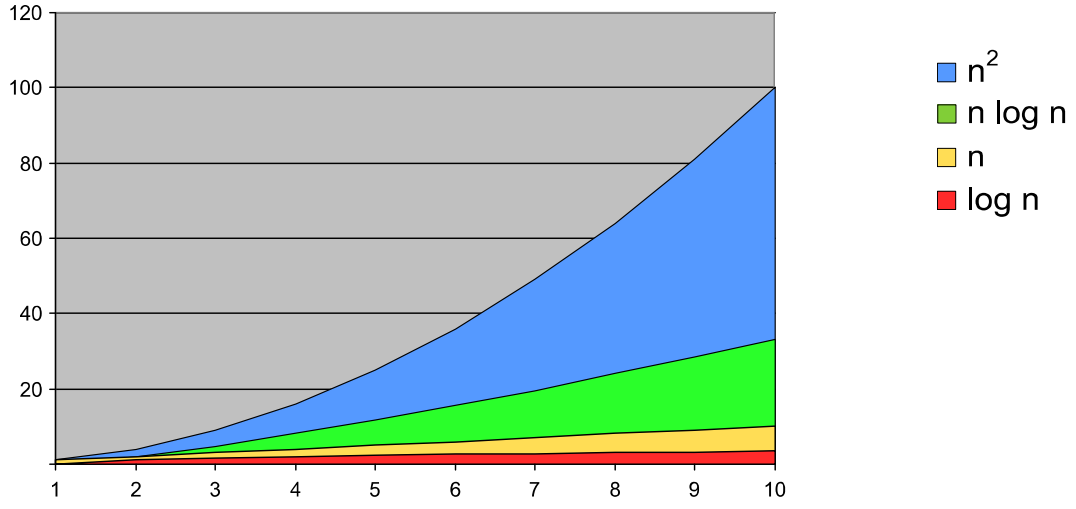
$O(n^2)$

- Represents an algorithm that has "quadratic" growth
- These algorithms grow dramatically fast depending on the size of n
- **Do <u>NOT</u> use for large values of n**

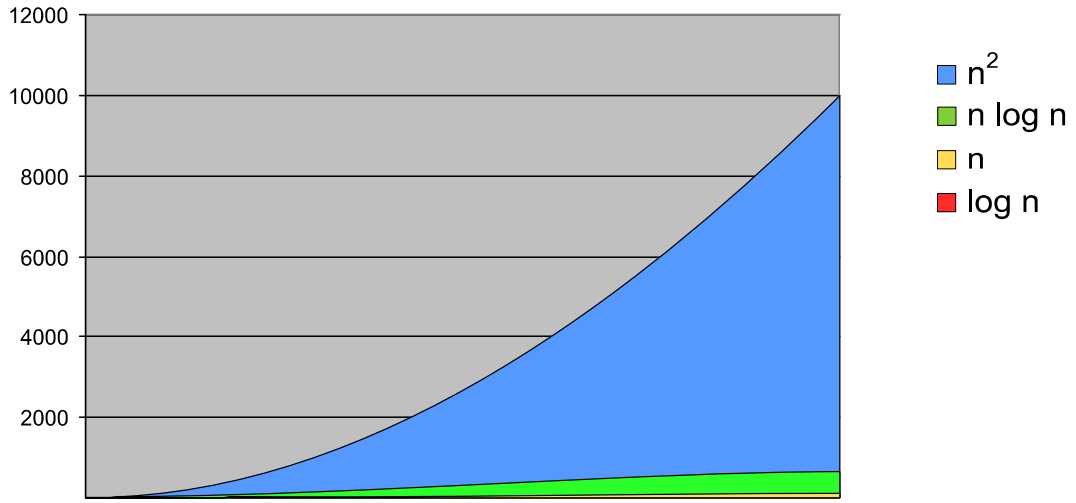
$O(n^2)$ Examples

- Bubble Sort, Selection Sort, etc….
- matrix multiplication
- merging unsorted arrays

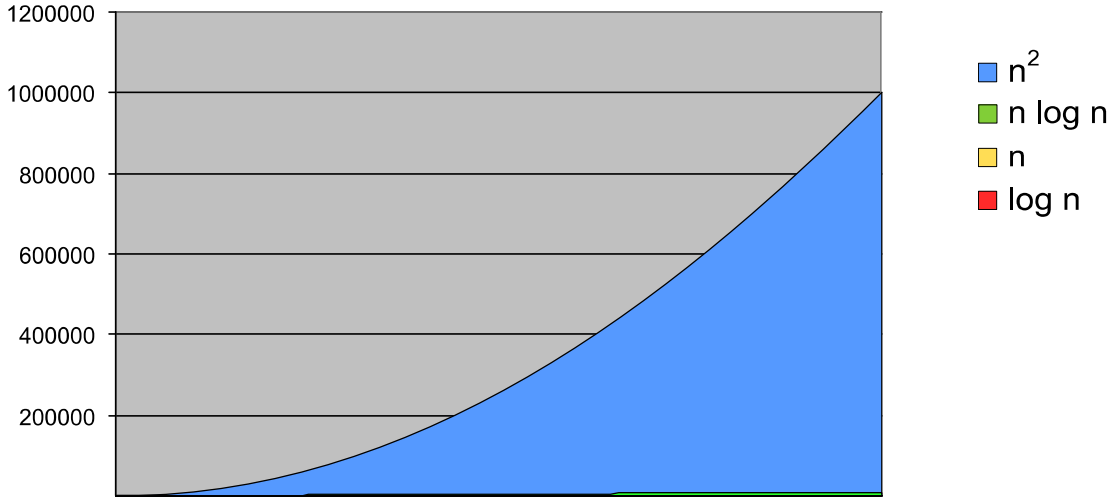
Growth: 1 to 10



Growth: 1 to 100



Growth: 1 to 1000



Seconds needed (1 Microsecond Operation)

n	O(log n)	O(n)	O(n log n)	O(n^2)
10	0.000003	0.000010	0.000033	0.000100
100	0.000007	0.000100	0.000664	0.010000
1,000	0.000010	0.001000	0.009966	1.000000
10,000	0.000013	0.010000	0.132877	100.000000
100,000	0.000017	0.100000	1.660964	2.8 hours
1,000,000	0.000020	1.000000	19.931569	11.6 days
10,000,000	0.000023	10.000000	3.9 minutes	3.16 years
100,000,000	0.000027	100.000000	44.3 minutes	316.9 years
1,000,000,000	0.000030	17 minutes	8.3 hours	31,689 years

Big-O Summary

Good	O(1)	O(wow!)
	O(log n)	O(yeah!)
	O(n)	O(nice!)
	O(n log n)	O(kay!)
	O(n^2)	O(no!)
Bad	O(2^n)	O(sh*t!)

Big-O Math

Time for a "Big-O" headache

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Asymptotic Analysis

- Any algorithm can be analyzed, and its complexity/growth can be written as a simple mathematical expression
- *Asymptotic analysis* of an algorithm determines the running time in big-O notation

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Asymptotic Analysis

1. Find the worst-case number of primitive operations executed as a function of the input size
2. Eliminate meaningless values once the base rate in found

Example

- If we analyze an algorithm and find it executes **12 * n – 1**
  - constant factors and lower-order terms dropped since they become meaningless for large values of n
  - remember, this is a *growth rate*
- It will be **O(n)**

Examples

- $3000n + 7 \rightarrow O(n)$
- $2n^5 + 3n^3 + 5 \rightarrow O(n^5)$
- $7n^3 - 2n + 3 \rightarrow O(n^3)$

Test Your Might…

```
for(i = 0; i < 100; i++)
{
   total += values[i];
}
```

O(1)

Test Your Might…

```
for(x = 0; x < array.size; x++)
{
     sum += array[x];
}

for(x = 0; x < array.size; x++)
{
     sum -= array[x];
}
```

O(n)

Test Your Might…

```
for (x = 0; x < array.size; x++)
{
    for (y = 0; y < x; y++)
    {
        sum += x - y;
    }
}
```

$O(n^2)$